

Sir Clive Sinclair's super-fast Risc pops up

Apart from designing a briefcase bicycle, Sir Clive Sinclair has been remarkably quiet in his back room since he sold off his interest in Cambridge Computer last year; one of the products he was working on has at last popped its head up in public.

The product is a super-fast Risc, called the PpC (Pretty good Computer?) and what it is good at (according to Sinclair) is emulating other machines.

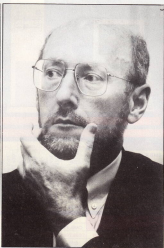
Getting hard information about it has been almost impossible; but it is now fully under the control of Sinclair's colleague Chris Shelton, who operates from a new computer company, PpC Limited. And whether this is the reason I can't say, but suddenly, full specifications are available.

According to *Electronic World News*, which was given an 'unofficial leak' by official sources, the machine will be able to run 80286 instructions as fast as a 12MHz 286. That, of course, is not news; what is news is the detail of how it is done.

The *EWN* article says that the PpC 7600 'treats machine code as data, and translates the data into its own reduced instruction set'.

Data: the machine is 'self-clocking' — it doesn't wait for an external signal, but triggers an instruction when the previous instruction ends, making it have an 'equivalent clock' speed of 200MHz.

There's a bank of 40 registers on chip, each 32 bits wide. Instructions are fed into



a pre-fetch cache with room for 32 instructions, and there are special areas of the chip where ROM instructions can be built in to target a specific architecture.

And where is this miracle? Well, when it is available, it will be made by Plessey, which takes us a whole step nearer reality. But it still isn't available.

What makes the *EWN*

Sir Clive busy inventing super-fast Risc is the back room.

article interesting is the suggestion that Sinclair is proposing to make the chip part of his wafer-scale Anamorphic research.

The PpC chip uses video RAM — dual-port RAM, which can be written from one processor while being read by another. This, plus other high-

speed links between the processor in a very Transputer-like way, means that multi-processor machines can be built to run very, very fast.

Put the chips all on the same wafer, as Sinclair is now suggesting, and the speed could become enormous.

Alas, no silicon has yet been built. Money, I assume, is the reason. 'We're getting closer to silicon, I believe,' was as far as Uncle Clive himself would commit himself when I asked him about it. But apart from this year, probably he wouldn't be more specific. And Shelton's answering machine was even less forthcoming.

The basic processor is about 100,000 transistors, not large by today's standards. It's going to be built in a low-power silicon technology, and should be ideal for portable use, but the market for portable computers is changing rapidly. The 386SX is now commonplace, and it's hard to see what you'd get from an emulation of the 286.

What none of the notebook players have built is a proper communicating laptop: Hewlett-Packard's Palmtop is the closest so far. I'll see if I can't inspire Sinclair to think radio comms... next time I'm trying out his bicycle.

Shelton himself was one of the founders of the UK computer industry with a remarkable bit of multi-processing called the Sig-Net, which actually sold quite well in the days of CP/M.

Kodak's Diconix M150 Plus comes down in price

Kodak's Mac compatible portable ink-jet printer, the Diconix M150 Plus, is down in price to £399; it offers a draft speed of 7.5m/sec, quality script, plain paper printing, mains or rechargeable battery supply and an operating noise of 45dB. Adobe Type Manager (ATM) comes as standard plus graphics support software and all 13 PostScript outline fonts provided with the original Apple LaserWriter. The unit weighs 3/1 lb and measures 1.97 x 6.5 x 10.8 in.

If you've recently acquired one of the new low-price Macintosh range and you're looking for something on which to print all your impressive new output, you could take a look at this.

Kodak is on 0442 61122.

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